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PROVISIONAL PATENT SPECIFICATION

SYSTEM AND METHOD FOR A BIDIRECTIONAL, SELF-HEALING ACTIVE-MATRIX OPTICAL CONNECTOR AND FREE-SPACE MOBILE DATA EXCHANGE LAYER

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PROVISIONAL PATENT DISCLOSURE

FIELD OF THE INVENTION

This invention relates generally to telecommunications, hardware interfaces, mobile computing, and optical data transmission. More specifically, this invention relates to a bidirectional, self-healing active-matrix optical interface that supports both (1) guided, interlocking cable connector couplings (JADC2-CVBNI), and (2) unguided, free-space open-air optical data exchange between uncoupled mobile devices, utilizing co-deposited light-emitting and light-sensing subpixels, de-phased half-duplex duty cycles, and real-time active-matrix coordinate alignment mapping.

BACKGROUND OF THE INVENTION AND TECHNICAL CHALLENGE

Data transmission over short distances is traditionally limited by physical, electrical, and environmental constraints.

First, in high-bandwidth guided interconnects—such as the optical data cable of Hub Application No. 19/559,705—the communication link is strictly unidirectional and asymmetrical. One side of the coupler houses the emitters (micro-display) and the other houses the sensors (image sensor). To achieve bidirectional, full-duplex communication, legacy systems must run separate, parallel transmit (TX) and receive (RX) lines, doubling the physical footprint of the connectors, increasing thermal output, and complicating cable management in high-density defense and datacenter environments.

Second, in unguided free-space optical exchanges—such as the short-range mobile data link of Application No. 19/560,042—devices utilize the screen as an emitter and the camera as a receiver. This setup requires the users to maintain rigid, near-perfect physical alignment between the screen of Device A and the camera of Device B. However, in real-world mobile environments, physical movement, hand-shake, and rotational skew disrupt the optical line of sight, causing high packet drop and link failure. This makes ad-hoc data exchange between uncoupled devices highly unreliable without bulky docking stations, jigs, or static stands.

Furthermore, high-speed optical systems experience physical signal degradation, including chromatic cross-talk (color bleeding), dispersion, and physical alignment drift, which restrict data fidelity.

A unified, bidirectional, self-aligning, and mechanically resilient optical communication system is needed that can operate over both guided (connectors) and unguided (free-space open-air) mediums, dynamically compensating for physical alignment shift in real-time while delivering data natively at the speed of light.

THE BIDIRECTIONAL ACTIVE-MATRIX OPTICAL COUPLER (JADC2-CVBNI)

The present invention solves these problems by providing a Bidirectional, Self-Healing Active-Matrix Optical Connector and Free-Space Mobile Data Exchange Layer.

In a first primary embodiment (the guided connector coupling), the system merges the unidirectional optical cable of Application No. 19/559,705 with the monolithic bidirectional display of Application No. 19/746,749 (P.E.R.W.I.N.).

Both connector heads in the JADC2-CVBNI interface house a common, monolithic co-deposited active-matrix plane containing interleaved light-emitting subpixels (OLED/QDEL) and organic

photodetector (OPD) sensor subpixels.

To achieve full bidirectional communication over a single, unified interface without display-to-sensor glare, the system executes a high-speed, synchronized, and de-phased half-duplex duty cycle. During phase one, Connector Head A's light-emitting subpixels are active (emitting), while Connector Head B's photodetector subpixels are active (sensing). During phase two, the states reverse. Because this duty cycle operates at megahertz or gigahertz frequencies, the physical interface achieves continuous, zero-latency bidirectional data flow.

REAL-TIME ACTIVE-MATRIX COORDINATE ALIGNMENT MAPPING

To completely eliminate the requirement for precision mechanical tolerances and alignment jigs, the system integrates the self-healing coordinate anchoring of Application No. 64/126,057.

Upon physical mating or during open-air free-space transmission, the receiving sensor array executes a real-time coordinate calibration sweep. The sensor array scans the incoming visual data matrix to detect the exact, offset spatial coordinates of the emitting pixels. The integrated controller ASIC/FPGA then dynamically maps the logical data channels directly to the detected physical coordinates.

If the connectors are physically jarred, or if a user's hand shakes during free-space transmission, the active-matrix controller instantly shifts its coordinate map to track the movement in real-time. Even if minor alignment drift causes some pixels to lose line-of-sight (resulting in a minor, temporary bandwidth reduction), the communication link remains **unbroken**, routing packets around the dead or misaligned coordinates.

FREE-SPACE BIDIRECTIONAL MOBILE DATA EXCHANGE

In a second primary embodiment (the unguided free-space mobile link), the exact same active-matrix bidirectional communication and self-healing alignment technologies are deployed over open air between uncoupled mobile devices (such as smartphones or tablets).

If the displays of both mobile devices are manufactured using the P.E.R.W.I.N. bidirectional substrate, the devices can establish a high-speed, bidirectional free-space link simply by pointing their screens at each other over open air.

Using the self-healing coordinate mapping, the devices track each other's screen movements and angular skew in real-time, completely compensating for standard hand-shake or physical displacement. This allows users to hold their smartphones loosely and transfer gigabytes of data natively and securely out-of-band at the speed of light, completely bypassed from legacy cellular or Wi-Fi networks.

ISOLATED FACE-TO-FACE MOBILE COUPLING SHELL

To establish an incredibly robust, secure, and low-cost consumer-grade version of the bidirectional optical data link, the system utilizes a passive, light-blocking mechanical coupling shell.

The coupling shell is a physical, un-powered sleeve or housing designed to hold a first mobile device and a second mobile device in a rigid, face-to-face configuration where their display screens are aligned in close proximity. The shell completely seals the aligned displays from all external ambient light, establishing an isolated dark chamber.

By eliminating all external ambient glare and environmental interference inside this dark chamber, the co-deposited OLED emitters and OPD sensors on the screens (per 19/746,749) can operate at maximum signal-to-noise ratios (SNR), enabling ultra-high-bandwidth, zero-interference, and

multi-gigabit bidirectional data exchange over open air with zero physical cables. This allows users to securely migrate the entire contents of a smartphone to a new device in seconds, out-of-band, without any network dependency.

Furthermore, this face-to-face coupling shell enables the establishment of a secure, out-of-band, and completely air-gapped bidirectional optical communication link between two independent computers (which may be classical or optical computing systems). In this configuration, a first computer is connected via a standard interface (such as USB, Thunderbolt, or a hardware bus) to the first mobile device, and a second computer is connected similarly to the second mobile device. The two mobile devices, when placed face-to-face inside the light-blocking coupling shell, act as active transceiver couplers (optical-to-electrical-to-optical bridges).

Data transmitted from the first computer is translated by the first mobile device into high-radix visual frames, emitted across the unguided dark chamber, captured by the second mobile device's OPD sensor array, and decoded back into data delivered directly to the second computer, achieving a highly secure, cable-free, and EMP-proof server-to-server data-link layer using consumer-grade mobile transceivers.

BRIEF DESCRIPTION OF THE DRAWINGS

To complement this description and support understanding of the visual components, standard-compliant drawings are attached on separate sheets.

- **FIG. 1** is an exploded, conceptual diagram of the guided bidirectional active-matrix connector head (JADC2-CVBNI), showing the co-deposited subpixel active plane.
- **FIG. 2** is a conceptual schematic of two uncoupled mobile devices establishing a free-space, open-air optical data link over open air.

- **FIG. 3** is a perspective view of the mechanical/optical coupling shell holding two mobile devices face-to-face, completely sealing them from external ambient light.
- **FIG. 4** is a system diagram illustrating the computer-to-computer optical bridge configuration, where two external host computers are connected respectively to the two mobile devices, which act as optical transceiver couplers inside the coupling shell.

SOFTWARE-DEFINED ACTIVE OPTICAL LOGIC GATES

Because both sides of the JADC2-CVBNI interface (or both mobile screens) are active emission and detection planes, the integrated controller ASICs can be programmed to execute arbitrary, software-defined optical logic rules directly on the visual matrix in transit:

$$\textbf{Red (\#FF0000)} \oplus \textbf{Blue (\#0000FF)} \;\; \textbf{Purple (\#AF23CC)}$$

This enables the interface itself to function as a massively parallel, software-defined optical logic switch, executing cryptographic transformations, phase-rotations, and rotating encoding maps natively on the physical light stream before demodulation occurs.

CLAIMS

What is claimed is:

1. A bidirectional, self-healing active-matrix optical data transmission system comprising:

    - a first active-matrix substrate and a second active-matrix substrate, wherein each substrate supports an interleaved array of co-deposited light-emitting subpixels and light-receiving photodetector subpixels;
    - a timing generator configured to execute a de-phased half-duplex duty cycle, wherein light-emitting subpixels on said first substrate transmit data frames during a period when the photodetector subpixels on said second substrate are actively integrating, and wherein states reverse on a subsequent phase; and
    - an active-matrix controller configured to execute a real-time coordinate calibration sweep to map and track the spatial coordinates of emitting pixels on said photodetector subpixels, dynamically shifting the data-link coordinate map in response to physical offset, vibration, or movement to maintain an unbroken communication link.

2. The system of Claim 1, wherein the light-emitting subpixels are organic light-emitting diodes (OLED) or quantum dot electroluminescent (QDEL) diodes, and the photodetector subpixels are organic photodetectors (OPD).

3. The system of Claim 1, wherein the first substrate and the second substrate are housed within interlocking connector heads of a physical data cable to establish a guided, bidirectional optical interface.

4. The system of Claim 1, wherein the first substrate and the second substrate are the display screens of uncoupled mobile electronic devices, establishing a free-space, bidirectional open-air optical communication link.

5. The system of Claim 4, wherein said active-matrix controller dynamically compensates for physical hand-shake or displacement between the uncoupled mobile devices in real-time.

6. The system of Claim 1, further comprising an integrated processor configured to execute software-defined active optical logic rules on the passing light stream, performing logical operations on the colors and intensities of the transmitted pixels before demodulation.

7. A mechanical coupling shell for a bidirectional optical data transmission system, comprising a physical housing configured to hold a first mobile electronic device and a second mobile electronic device in a rigid, face-to-face configuration, wherein said physical housing completely seals said display screens from external ambient light, establishing an isolated, zero-interference dark chamber for bidirectional, free-space open-air optical data exchange between the uncoupled devices.

8. The coupling shell of Claim 7, wherein said display screens are manufactured using the bidirectional active-matrix substrate of Claim 1.

9. The system of Claim 4, wherein said first mobile electronic device is connected to a first computer and said second mobile electronic device is connected to a second computer, and wherein said first and second mobile electronic devices act as optical transceiver couplers to establish a secure, out-of-band and air-gapped bidirectional data link between said first and second computers.

ABSTRACT OF THE DISCLOSURE

A bidirectional, self-healing active-matrix optical data transmission system and method are disclosed. The system integrates display emission and light capture over a unified interface, supporting both guided cable connector couplings (JADC2-CVBNI) and unguided free-space open-air optical data exchange between uncoupled mobile devices. Both interface planes comprise a monolithic, co-deposited array of interleaved light-emitting subpixels and organic photodetectors. Communication operates via a high-speed, synchronized, and de-phased half-duplex duty cycle. To eliminate precision mechanical tolerances and alignment constraints, an active-matrix controller executes a real-time coordinate calibration sweep, mapping and tracking the spatial coordinates of emitting pixels. If the interface experience physical offset, vibration, or hand-shake during open-air transmission, the controller dynamically shifts its coordinate map to maintain an unbroken, zero-latency optical communication link.